Bishnu Datta Pant

Email: bishnupant92@gmail.com
Mobile: +977 9865376807
GitHub: LINK

RESEARCH INTERESTS

Precision agriculture, remote sensing and satellite imagery analysis, AI/ML applications in agricultural systems, UAV data processing, land use/land cover classification, and geospatial data science for food security in South Asia and Africa.

EDUCATION

Gokuleswor Agricultural and Animal Science College, Tribhuvan University, *Baitadi, Nepal*
Bachelor of Science in Agriculture (B.Sc. Ag) | Final Grade: 68.95% | 2019 – March 2023

WORK EXPERIENCE

Agricultural Technology Centre Pvt. Ltd. (ISO/IEC 17025:2017 Accredited) *Lalitpur, Nepal*
Laboratory & Marketing Assistant | July 2025 – March 2026

- **Soil & Plant Analysis:** Nutrient profiling (N, P, K, micronutrients) on samples using accredited protocols; maintained QC documentation.

Agricultural Technology Centre Pvt. Ltd. (ISO/IEC 17025:2017 Accredited) *Lalitpur, Nepal*
Laboratory & Marketing Assistant | July 2025 – March 2026

- Designed engaging posters and promotional content
- Managed social media platforms and organized online workshop on GIS and Remote Sensing

VOLUNTEER EXPERIENCE

Climate Geodesign of Lagos Team, *Remote* / *Lagos, Nigeria*
Volunteer GIS Contributor – Global Climate Geodesign Competition | April 2025 – Present

LULC Mapping: Produced Land Use and Land Cover classification map for Lagos State, Nigeria (2025) using QGIS and satellite imagery; contributed spatial datasets to the team's climate resilience planning repository.

International Collaboration: Worked with a multinational team on climate geodesign scenarios; integrated geospatial outputs into urban resilience and land planning frameworks

RESEARCH & TECHNICAL PROJECTS

- **Machine Learning Crop Yield Prediction:** Predictive models using agronomic and meteorological datasets; feature engineering and cross-validation with Scikit-learn.
- **Crop Recommendation System:** Multi-class classification using soil nutrients (N, P, K, pH, Zn, S), weather variables; Random Forest and SVM applied across 22 crop classes.
- **LULC Classification – Lagos State, Nigeria:** Supervised satellite imagery classification in QGIS; produced validated land cover map for 2026 Climate Action plan drawing of Lagos state for GCGC app

PUBLICATIONS

- Khanal et al. (2025). Evaluation of Rice Landraces for Yield and Related Traits Under Rainfed Conditions in Nepal. *AgroEnvironmental Sustainability, 3(2), 130–139.*
- Khanal et al. (2025). Economic Perspectives on Walnut Production and Trade in Darchula District of Nepal. *Asian Journal of Advances in Agricultural Research, 25(5), 123–129.*
- Bist et al. (2025). Nanotechnology in Agriculture: A Review of Innovations in Crop Protection and Food Security. *Advances in Agriculture, Article ID 8892001 (Wiley).*

SKILLS

- **Programming & Data Science:** Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn)
- **Machine Learning & AI:** Scikit-learn, PyTorch (learning), crop yield prediction, multi-class classification, feature engineering, model validation
- **Geospatial & Remote Sensing:** QGIS (Advanced), ArcGIS, Google Earth Engine (Begeinner), LULC classification, spectral indices, georeferencing, digitization, satellite imagery analysis
- **Laboratory & Field:** Soil & plant tissue analysis (ISO/IEC 17025:2017), crop diagnostics, field experimental design (Alpha-Lattice, RCBD)

CERTIFICATIONS

- **QGIS for Land Management** – Centre of Excellence in Land Administration and Management (ATI, India)
- **Data Science with Python** – Evolve IT Hub , Nepal
- **Introduction to R** – Great Learning Academy

LANGUAGES

- **Nepali:** Native
- **English:** B2 – Professional working proficiency